

SWASTI PAHUJA

Applied ML Engineer | Credit Risk + GenAI | IIT Bombay 2024

 swastipahuja@gmail.com  linkedin.com/in/swasti-pahuja2002/  swastipahuja2002.github.io/  +91 7023557551

Education

B.Tech, Civil Engineering | Indian Institute of Technology, Bombay (IIT Bombay)

Nov'20 – May'24

CGPA: 7.39 / 10 | Relevant Coursework: Probability & Statistics, Calculus, Differential Equations, Computer Programming, Economics

- JEE Advanced 2020: Top 3.9% of ~150,000 candidates | JEE Main 2020: 99.4 percentile | CBSE Class XII: 96%

Professional Experience

IDFC FIRST Bank | Data Scientist

Jul'24 – Present

Pre-Placement Offer post-internship in Credit Risk & Analytics; built application + behavioral models across secured retail lending

- Used Car Refinance Banking Application Scorecard

- Built and validated logistic regression pipeline (univariate, WoE/IV, inter/intra-cluster variable clustering, step down logit); achieved **38%** Gini validation, and replaced the legacy model with 6X lift in discriminatory power on out-of-time sample
- Among the first ML models deployed in the bank to leverage in-house BSA over third-party Perfios; delivered **~₹3 Cr** annual cost savings across **~50K** monthly applications by migrating scoring pipeline along with in-house BSA infrastructure

- Two-Wheeler Used Repurchase: Dual ML Scorecard (KGB + AGB)

- Introduced first segment-specific ML scorecard for **₹105 Cr/month** Two-Wheeler portfolio, replacing generic commuter model
- Built parallel models for the KGB via logistic regression and AGB via XGBoost (RFECV, BorutaSHAP, Optuna); delivered **~1.5X** Gini lift over the commuter-scorecard baseline across both KGB and AGB approval stages on out-of-time validation
- Implemented reject inferencing via the multiplier method; projected NCL reduction from **4.6%** to **<3.8%** post-deployment

- ML Productionization & Cross-Portfolio

- Productionized logistic regression frameworks after Hosmer–Lemeshow calibration for stable 10-decile risk ranking; chose interpretable LR over higher AUC XGBoost, a deliberate accuracy-vs-deployability trade-off given 10 decile requirement
- Automated model monitoring via Airflow, reducing manual latency by **~60%** across PSI, CSI and decile level shift in scorecards

Projects

AI-Powered Local File Search (RAG System) | Personal Project

Apr'26 – Present

Local-first semantic search and Q&A engine over a personal document corpus across PDFs, docs, and notes

- Built RAG pipeline from scratch: text extraction across 5 file formats, recursive chunking with overlap, OpenAI text-embedding 3-small vectors, cosine-similarity retrieval via ChromaDB, grounded generation with source attribution
- Refactored from-scratch implementation onto LlamaIndex with persistent ChromaDB for incremental cross-session indexing
- Built content-hash-based incremental indexing; building Streamlit interface with metadata filtering and source-cited answers

AssembleAI - Voice Based RAG Assistant

Nov'25 – Present

Voice assistant that answers assembly questions from uploaded product manuals, hands-free guidance during physical setup tasks

- Developed voice assistant grounding spoken responses in user-uploaded assembly manuals using ElevenLabs and LlamaIndex
- Won Bolt and Clerk sponsor awards at the ElevenLabs Hackathon, Mumbai, placed **4th** overall among 15+ teams

NEXRAD Radar Signal Classification | Bachelor Thesis, IIT Bombay (Prof. J. Indu)

Aug'23 – May'24

Built ML pipeline classifying meteorological vs. biological radar signals at scale on NEXRAD data

- Trained classification model on NEXRAD radar data; achieved **~90%** accuracy across heterogeneous radar regimes
- Implemented automated radar signal processing in Python; presented at **EGU General Assembly 2024**, Vienna, Austria, EU

Skills

Machine Learning & Statistics: Logistic Regression, XGBoost, Gradient Boosting, Feature Engineering, BorutaPy, BorutaSHAP, RFECV, Optuna, WoE/IV, CSI, Hierarchical Clustering, Hosmer–Lemeshow Calibration, Reject Inferencing, SHAP, PSI/CSI monitoring

GenAI & LLMs: RAG, LlamaIndex, ChromaDB, Vector Databases, Semantic Search, LLM Evaluation, Embedding Models

Programming & Data: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn), PySpark, Streamlit, Bash, Excel (Advanced)

Productionization & Tools: Apache Airflow, Streamlit, ChromaDB, Git, AWS (S3, EC2, SageMaker), Cursor, Claude Code

Domain: Credit Risk Modeling, Banking Scorecards, Behavioral Analytics, Portfolio Risk, Underwriting, Model Monitoring

Leadership and Achievements

- Overall Coordinator, Izhaar Festival, IIT Bombay (2023): Led 10+ team members; 800+ attendee footfall; ₹2L+ budget
- Mentor, Summer of Science, IIT Bombay (2022): Guided 4 mentees through 3-month project on Stock Market Analysis
- Web Secretary, Civil Engineering Association, IIT Bombay (2021–22): Redesigned departmental website serving 900+ students
- 1st place, Abhisaran International Radioplay (2021) | 3rd place, Badminton Girls GC, IIT Bombay (2023)